THE KISSING NUMBERS IN DIMENSIONS FIVE, SIX AND SEVEN ARE AT MOST 43, 76 AND 132

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ABSTRACT. The kissing number τ_n is the largest number of non-overlapping unit balls in $\mathbb{R}^n$ that can touch a common unit ball. The previous best bounds were $40 \le \tau_5 \le 44$, $72 \le \tau_6 \le 77$ and $126 \le \tau_7 \le 134$. The upper bounds for τ_5 and τ_7 come from Mittelmann and Vallentin's high-precision floating-point computations of the three-point bound; the upper bound for τ_6 comes from the second level of the Lasserre hierarchy for spherical codes, as computed by de Laat, Leijenhorst and de Muinck Keizer. We prove $\tau_5 \le 43$, $\tau_6 \le 76$ and $\tau_7 \le 132$. Each proof is a certificate: an exact rational solution of the second level of the hierarchy at degrees $(d_1, d_2, \delta) = (14, 20, 20)$, in the formulation of de Laat, Leijenhorst and de Muinck Keizer. We check each certificate with the verification program of these authors. Other exact certificates for the same hierarchy give upper bounds on the largest size $A(n, \arccos \frac{1}{3})$ of a spherical code in $\mathbb{R}^n$ with minimal angle $\arccos \frac{1}{3}$: $A(4, \arccos \frac{1}{3}) \le 14$, $A(5, \arccos \frac{1}{3}) \le 22$ and $A(6, \arccos \frac{1}{3}) \le 34$, improving the previous bounds 15, 23 and 35; since a 14-point code in $\mathbb{R}^4$ is known, $A(4, \arccos \frac{1}{3}) = 14$. The number $A(n-1, \arccos \frac{1}{3})$ is also the largest number of points of a kissing configuration in $\mathbb{R}^n$ that lie at angle $\pi/3$ from one point of the configuration, so in $\mathbb{R}^5$ this number is $A(4, \arccos \frac{1}{3}) = 14$. We also prove our six upper bounds twice in the Lean proof assistant, once with compiled code and once with every computation done by Lean's kernel; both proofs assume only the positivity of the zonal kernels given by zonal polynomial files.

1. INTRODUCTION

A kissing configuration in $\mathbb{R}^n$ is a set $C \subset S^{n-1}$ with $\langle x, y \rangle \le \frac{1}{2}$ for all distinct $x, y \in C$, and the kissing number τ_n is the largest size of such a set. The values $\tau_1 = 2$ and $\tau_2 = 6$ are elementary; $\tau_3 = 12$ is due to Schütte and van der Waerden [22]; $\tau_4 = 24$ is due to Musin [20]; $\tau_8 = 240$ and $\tau_{24} = 196560$ are due to Odlyzko and Sloane [21] and, independently, to Levenshtein [17]. In dimension five the root system D_5 gives $\tau_5 \ge 40$ (Korkine and Zolotareff [11]), and three further non-isometric 40-point kissing configurations are known, found by Leech [15], by Szöllősi [24] and by Cohn and Rajagopal [5]. The linear programming bound yields $\tau_5 \le 46$ [21], and the three-point semidefinite programming bound of Bachoc and Vallentin [1] improves the bound to $\tau_5 \le 45$. Mittelmann and Vallentin [19] computed the three-point bound in high-precision floating-point arithmetic and obtained $44.99899685\ldots$, so $\tau_5 \le 44$. Three-point computations at higher degrees have since reached 44.981067 [18] and 44.970252 [16]. In dimensions six and seven the root systems E_6 and E_7 are kissing configurations with 72 and 126 points, and the three-point bound gives $\tau_6 \le 78$ and $\tau_7 \le 135$ [1]. For $n = 7$ the computation of [19] reaches $134.4488169\ldots$, so $\tau_7 \le 134$. De Laat, Leijenhorst and de Muinck Keizer [13] computed the second level of the Lasserre hierarchy at degrees $(14, 16, 16)$ in dimensions 5, 6, 7, 10, 12 and 16. This computation improved the known bounds only in dimension six, to $\tau_6 \le 77$. See [28] for a recent survey.

Theorem 1.1. *$\tau_5 \le 43$, $\tau_6 \le 76$ and $\tau_7 \le 132$.*

More generally, a spherical code with minimal angle θ is a finite set $C \subset S^{n-1}$ with $\langle x, y \rangle \le \cos\theta$ for all distinct $x, y \in C$, and $A(n, \theta)$ denotes the largest size of such a code; thus $\tau_n =$

$A(n, \pi/3)$. Upper bounds for $A(n, \theta)$ come from linear programming (Delsarte, Goethals and Seidel [6]; Kabatiansky and Levenshtein [10]) and from semidefinite programming. Bachoc and Vallentin [1] tabulated the three-point bound for several dimensions and angles. For $\cos\theta = \frac{1}{3}$ their table matched the bound $A(4, \theta) \le 15$, improved the upper bound for $A(5, \theta)$ from 24 to 23 and improved that for $A(6, \theta)$ from 37 to 35. The bounds $A(4, \theta) \le 15$ and $A(5, \theta) \le 24$ are due to Boyvalenkov and Kazakov [3]. The best known codes in dimensions 4, 5 and 6 have 14, 21 and 32 points (Section 7). (For $n = 5$ the table of [1] lists the weaker lower bound 20.) Before this work the known bounds were therefore

$$14 \le A(4, \theta) \le 15, \qquad 21 \le A(5, \theta) \le 23, \qquad 32 \le A(6, \theta) \le 35.$$

Cohn, de Laat and Leijenhorst [4, Table 6.1] list the 32-point code in $\mathbb{R}^6$ among the codes conjectured to be optimal.

Theorem 1.2. *Let $\theta = \arccos\frac{1}{3}$. Then $A(4, \theta) = 14$, $A(5, \theta) \le 22$ and $A(6, \theta) \le 34$.*

Together with the codes of Section 7, Theorem 1.2 yields $A(5, \theta) \in \{21, 22\}$ and $32 \le A(6, \theta) \le 34$.

For a kissing configuration $C \subset S^{n-1}$ and $x \in C$, call the points $y \in C$ with $\langle x, y\rangle = \frac{1}{2}$ the contacts of x. In the arrangement of unit balls centred at 0 and at $2y$ for $y \in C$, the contacts of x correspond to the outer balls that touch the ball centred at $2x$. Let c_n be the largest number of contacts of a point in a kissing configuration in $\mathbb{R}^n$.

Corollary 1.3. *For every $n \ge 2$ we have $c_n = A(n-1, \arccos\frac{1}{3})$. Hence $c_5 = 14$, $c_6 \in \{21, 22\}$ and $32 \le c_7 \le 34$.*

The angle $\arccos\frac{1}{3}$ enters through the tangent space at x: the normalised projections to $x^\perp$ of two contacts of x have inner product at most $\frac{1}{3}$, and conversely unit vectors in $x^\perp$ with pairwise inner products at most $\frac{1}{3}$ give contacts of x in a kissing configuration (Section 8). This classical correspondence is used, for example, in the proof of [7, Theorem 4.2]. With the bounds of [1] the correspondence implies $c_5 \le 15$, $c_6 \le 23$ and $c_7 \le 35$. Theorem 1.2 and Corollary 1.3 were first published in [25]. We include them here because they come from the same method as Theorem 1.1.

All upper bounds in this paper come from the second level of the Lasserre hierarchy for spherical codes. De Laat and Vallentin [12] defined the hierarchy, and de Laat, Leijenhorst and de Muinck Keizer [13] reduced its second level by symmetry. For each bound we solved the reduced semidefinite program numerically, rounded the solution to an exact rational solution and checked the result with the verification program of [13]. For τ_5, τ_6 and τ_7 we rounded a convex combination of two dual iterates of the interior-point method. The two iterates have values on either side of a target value (Section 4). Section 2 recalls what a certificate consists of. Sections 3–5 treat τ_5, τ_6 and τ_7: the floating-point values at increasing degrees, how the certificates were obtained, and how they were checked. Section 6 lists the certificates for $\cos\theta = \frac{1}{3}$, Section 7 recalls the lower bounds, Section 8 proves Corollary 1.3, and Section 9 describes Lean proofs of the upper bounds in Theorems 1.1 and 1.2. The Lean proofs assume that the zonal kernels are positive semidefinite.

2. The second level of the Lasserre hierarchy

We use the second level of the Lasserre hierarchy for spherical codes [12] in the symmetry-reduced form of [13, Section 4], and refer to that section for details. Fix n, $t = \cos\theta$ and the degrees (d_1, d_2, δ). Call a set of points of S^{n-1} admissible if their pairwise inner products lie in $[-1, t]$. The dual program asks for a positive semidefinite, orthogonally invariant kernel K

on the admissible sets of at most two points. After the symmetry reduction, K is given by the zonal matrices of [13] combined with positive semidefinite matrices F_λ, one for each partition $\lambda = (\lambda_1, \lambda_2)$ that occurs and has $|\lambda| \le d_1$. For $k = 1, 2, 3, 4$ and every admissible set S of k points, the sum of $K(J_1, J_2)$ over all J_1, J_2 with $J_1 \cup J_2 = S$ must be at most -1 if $k = 1$ and at most 0 otherwise. These inequalities are imposed as polynomial identities in the inner products of S: the difference between the two sides is a sum of terms $c\,g\,r$, where g is 1 or one of the symmetrised polynomials that cut out the admissible Gram matrices, $c \ge 0$ is a constant, r is a sum of squares, and each term has degree at most δ. We call the products $c\,g$ multipliers and refer to the requirements on c and g as the multiplier conditions. Every code C with minimal angle θ then has at most $K(\emptyset, \emptyset)$ points: since K is positive semidefinite, summing $K(J_1, J_2)$ over all subsets J_1, J_2 of C with at most two points gives a nonnegative number, and grouping the terms by $S = J_1 \cup J_2$ bounds this number by $K(\emptyset, \emptyset) - |C|$. The kernel K is positive semidefinite because the matrices F_λ are positive semidefinite and the zonal matrices of [13] are Gram matrices. The value $K(\emptyset, \emptyset)$ is one entry of the matrix F_λ for the trivial representation. The degrees $d_1 \le d_2$ determine the partitions λ and the sizes of the matrices F_λ. The even number $\delta \ge d_2$ bounds the degrees in the sums of squares.

A certificate consists of rational matrices F_λ and rational Gram matrices for the sums of squares, in fixed polynomial bases, such that all these matrices are positive semidefinite and all identities hold exactly. The verification program of [13] is written in Julia [2] with Nemo [8] and is available with the data of that paper [14]. It reads a certificate from files, checks that every matrix is positive definite by a Cholesky factorisation in ball arithmetic [9], transforms the matrices to the original problem, recomputes the four families of identities in exact rational arithmetic, and compares $K(\emptyset, \emptyset)$ with the claimed bound.

3. Floating-point values

Table 1 lists numerical values of the second-level relaxation for τ_5, τ_6 and τ_7 ($t = \frac{1}{2}$) at increasing degrees. These values are not part of the proof. We computed them in 128- or 256-bit floating-point arithmetic with a version of the interior-point solver of [16] that we had modified for speed. Raising d_1 from 14 to 16 at $\delta = 16$ lowers the value for τ_5 only slightly, from 44.3547 to 44.3511. At $d_1 = 14$, the steps of δ from 14 to 16, from 16 to 18 and from 18 to 20 lower the value for τ_5 by 0.70, 0.32 and 0.22, and lower the value for τ_7 by 3.00, 2.17 and 1.22. Raising δ from 16 to 20 at $d_1 = 14$ lowers the value for τ_6 by 1.26. The value for τ_7 at $\delta = 16$ is still above the three-point value 134.45, in line with [13].

(d_1, d_2, δ)	$n = 5$	$n = 6$	$n = 7$
$(10, 10, 10)$	46.3376	82.6312	
$(12, 12, 12)$	46.3052	82.4360	140.1624
$(14, 14, 14)$	45.0529		138.6072
$(14, 16, 16)$	44.3547	77.8418	135.6085
$(16, 16, 16)$	44.3511		
$(14, 18, 18)$	44.0314		133.4419
$(14, 20, 20)$	43.8154	76.5827	132.2202

Table 1. Dual objective at the end of the interior-point runs for the second level of the Lasserre hierarchy for τ_5, τ_6 and τ_7. Values are rounded to four decimals. Empty entries were not computed.

4. The certificates for τ_5, τ_6 and τ_7

In [13, Section 5.2], $\tau_6 \le 77$ was obtained by fixing the objective at 77.85, just above the numerical optimum. We likewise fix a target value: $N = \frac{4399}{100}$ for τ_5, $N = \frac{7699}{100}$ for τ_6 and $N = \frac{13299}{100}$ for τ_7. An exact rational solution of the relaxation with all matrices positive semidefinite and $K(\emptyset,\emptyset) = N$ proves $\tau_n \le \lfloor N \rfloor$. In each case we obtained such a solution by rounding a numerical one.

For τ_5 the interior-point method stalled when we applied it directly to the feasibility problem with $K(\emptyset,\emptyset) = N$, so for all three bounds we used two iterates of the optimization run instead. Let Y_A, Y_B be two dual iterates with values $b_A > N > b_B$. Consider the convex combination

$$W = (1-\gamma)\, Y_A + \gamma\, Y_B, \qquad \gamma = \frac{b_A - N}{b_A - b_B}.$$

Since the identities and $K(\emptyset,\emptyset)$ are linear in the matrices, W has $K(\emptyset,\emptyset) = N$ and satisfies all identities up to the residuals of Y_A and Y_B. It is positive definite because Y_A and Y_B are. We obtained Y_A and Y_B from two iterates of the run by one further step each in 256-bit arithmetic. This extra step reduces the dual residuals further. Table 2 lists the two iterates, their dual residuals before the extra step, the values b_A and b_B, and the smallest eigenvalue of W after each block is scaled to unit diagonal. For τ_5 and τ_7 the weight $1-\gamma$ is tiny, and W is essentially Y_B with its value raised to N. For τ_6 the value b_A is only 99.67, so the weight $1-\gamma$ is about 0.017. For τ_5 the extra step lowered the dual objective from the value 43.8154 in Table 1 to b_B. For τ_6, Y_B comes from iterate 126 of the 138 iterations of the run, so b_B lies above the final value in Table 1.

	τ_5	τ_6	τ_7
target N	4399/100	7699/100	13299/100
iterates giving Y_A and Y_B	96, 117	117, 126	99, 132
dual residuals of the iterates	$1\cdot10^{-17}$, $1\cdot10^{-18}$	$6\cdot10^{-19}$, $2\cdot10^{-19}$	$2\cdot10^{-17}$, $3\cdot10^{-18}$
b_A	$3.09\cdot10^{5}$	99.67	$3.06\cdot10^{9}$
b_B	43.8128	76.5965	132.2202
$1-\gamma$	$5.7\cdot10^{-7}$	$1.7\cdot10^{-2}$	$2.5\cdot10^{-10}$
smallest scaled eigenvalue of W	$8\cdot10^{-11}$	$9\cdot10^{-11}$	$1\cdot10^{-10}$
rounding	8 h (5 h)	11 h (4 h)	7 h (5 h)
size of the certificate files	2.3 GB	1.8 GB	2.1 GB
check: positive definiteness	1 min	1 min	1 min
check: transformation of the matrices	3 h	4 h	5 h
check: identities for $k \le 3$	28 min	11 min	14 min
check: identity for $k = 4$	23 h	32 h	37 h
check: peak memory	410 GB	730 GB	720 GB

Table 2. The certificates for τ_5, τ_6 and τ_7. Times are approximate wall-clock times. The rounding times are those of the run that produced the certificate. The part of each rounding time spent on the exact correction is given in parentheses. The times and the memory of the check are those of the run described in Section 5.

We rounded W with the procedure of Cohn, de Laat and Leijenhorst [4]: the entries are approximated by rationals, and a subset of the entries is corrected by an exact linear solve so that all identities hold exactly. In all three cases the identities amount to 11172 linear

equations in the 2666855 entries on and above the diagonals of the matrices. For τ_5 we allowed 255452 entries to change: all entries of the matrices F_λ and of the Gram matrices for $k \le 3$; the diagonal and superdiagonal of the Gram matrices for $k = 4$; and further entries chosen at random. For τ_7 this selection rule gave a system without a solution modulo two random primes, for every number of random entries we tried, up to 300140 entries in total. We then started over and added entries in batches of 100000 until the system had a solution modulo a random prime. The first entries added were those always selected by the rule, together with 45644 entries of the Gram matrices for $k = 4$ in positions whose two basis polynomials have the largest degree sum (20). The rest were random entries of the Gram matrices for $k = 4$. Two batches were needed. For τ_6 we used this batch procedure from the start, and eight batches were needed. In all three cases the correction is the minimum-norm solution restricted to 11731 of the allowed entries. All matrices stayed positive definite, so no change of basis was needed.

5. Verification of the certificates

All runs in this section use the package [14] without modification unless stated otherwise. We ran the verification program of [13] on the files in [26]. On the certificate for τ_5 the program reports that every spherical code in dimension 5 with pairwise cosine at most $\frac{1}{2}$ has at most $\frac{4399}{100}$ points. On the certificates for τ_6 and τ_7 it reports the same for dimension 6 with the bound $\frac{7699}{100}$ and for dimension 7 with the bound $\frac{13299}{100}$.[1] The checks read the zonal polynomial files of the supplement [26]. We computed these files with the package [14]. For $n = 5$, 6 and 7 we also recomputed the 490 files read by each check, starting from an empty cache, and obtained identical files. Since τ_5, τ_6 and τ_7 are integers, the three reported bounds prove Theorem 1.1.

Section 9 describes a second check of the certificates in Lean.

6. The certificates for $\cos\theta = \frac{1}{3}$

Table 3 lists the three certificates for Theorem 1.2. For each case we solved the semidefinite program numerically in multiple-precision arithmetic, with the modified solver of Section 3. The objective was replaced by the constraint $K(\emptyset, \emptyset) = N$ for the value N in the table. This value lies above the numerical optimum. The resulting feasibility problem then has solutions with positive definite matrices, and the interior-point method returns such a solution. We rounded this solution with the procedure of [4]. As with τ_5, τ_6 and τ_7, all matrices stayed positive definite.

n	previous bound	(d_1, d_2, δ)	level-2 value	certificate N	conclusion
4	15	$(12, 12, 12)$	14.9213	1499/100	$A(4, \theta) \le 14$
5	23	$(12, 12, 12)$	22.9177	459/20	$A(5, \theta) \le 22$
6	35	$(14, 16, 16)$	34.2444	349/10	$A(6, \theta) \le 34$

Table 3. Certificates for $\cos\theta = \frac{1}{3}$. The level-2 value is the floating-point optimum of the semidefinite program at the given degrees. Each certificate was checked by the verification program of [13]. The program reports that every code in $\mathbb{R}^n$ with minimal angle θ has at most N points.

[1] The output for τ_5 is "Verified that any spherical code with pairwise cosine at most 1//2 in dimension 5 contains at most 4399//100 points." The outputs for τ_6 and τ_7 are the same sentence with the dimension and the bound changed accordingly. We also ran a copy of the package, changed only to skip computing its auxiliary files (about 130 GB; see the README of [26]). This copy gave the same result on each certificate.

Since $A(n,\theta)$ is an integer, the certificates establish the upper bounds of Theorem 1.2.

Remark 6.1. We also computed the three-point bound of [1] for $n = 4$ in floating-point arithmetic. The table of [1] was computed at degree 10. At degree 12 the bound is 14.9903 and at degree 14 it is 14.9825, so at these degrees the three-point bound also points to $A(4,\theta) \le 14$. What we add in this case is an exact certificate. For $n = 5$ the second level at the degrees $(14, 16, 16)$ gives 22.4748, so these degrees do not improve the bound $A(5,\theta) \le 22$.

7. Lower bounds

The tables of [23] list codes of size 14 in $\mathbb{R}^4$ and of size 21 in $\mathbb{R}^5$ with minimal angle larger than $\arccos\frac{1}{3}$. In $\mathbb{R}^6$, the 32 unit vectors along 16 equiangular lines with angle $\arccos\frac{1}{3}$ [29] form a code of minimal angle $\arccos\frac{1}{3}$. For completeness the supplement [27] contains a 14-point code in S^3 and a 21-point code in S^4, each with rational coordinates and obtained by inverse stereographic projection of rational points. The largest pairwise inner products of these two codes are $0.32094\ldots$ and $0.31622\ldots$, both less than $\frac{1}{3}$. We checked both inequalities in exact arithmetic. Together with the certificate for $n = 4$, the 14-point code proves $A(4,\theta) = 14$.

8. Contact numbers

Proof of Corollary 1.3. Let $C \subset S^{n-1}$ be a kissing configuration and $x \in C$. Every contact y of x can be written as $y = \frac{1}{2}x + \frac{\sqrt{3}}{2}y'$ with $y' \in S^{n-1}$ orthogonal to x. For two distinct contacts y, z we get

$$\langle y, z\rangle = \tfrac{1}{4} + \tfrac{3}{4}\langle y', z'\rangle \le \tfrac{1}{2},$$

so $\langle y', z'\rangle \le \frac{1}{3}$. Hence the points y' form a code with minimal angle $\arccos\frac{1}{3}$ in the unit sphere of $x^\perp \cong \mathbb{R}^{n-1}$, and x has at most $A(n-1, \arccos\frac{1}{3})$ contacts. Conversely, let $x \in S^{n-1}$ and let D be a code with minimal angle $\arccos\frac{1}{3}$ in the unit sphere of $x^\perp$. The same computation shows that $\{x\} \cup \{\frac{1}{2}x + \frac{\sqrt{3}}{2}y' : y' \in D\}$ is a kissing configuration in which x has $|D|$ contacts. So $c_n = A(n-1, \arccos\frac{1}{3})$, and with the lower bounds of Section 7, Theorem 1.2 yields the values and bounds for $n = 5, 6, 7$. □

9. Lean proofs

We also checked the three certificates of Section 4 and the three certificates of Table 3 in the Lean 4 proof assistant with the Mathlib library. For $(n, t, M) = (5, \frac{1}{2}, 43)$, $(6, \frac{1}{2}, 76)$, $(7, \frac{1}{2}, 132)$, $(4, \frac{1}{3}, 14)$, $(5, \frac{1}{3}, 22)$ and $(6, \frac{1}{3}, 34)$, the matching Lean theorem states: if $\mathrm{ZonalPSD}(n, d_1, d_2)$ holds, then every finite set of unit vectors in $\mathbb{R}^n$ with pairwise inner products at most t has at most M elements. In this statement, d_1 and d_2 are the degrees of the certificate. The hypothesis $\mathrm{ZonalPSD}(n, d_1, d_2)$ says that for every partition λ that occurs and has $|\lambda| \le d_1$, the matrix kernel $(J, J') \mapsto \hat{Z}_\lambda(J, J')$ is positive semidefinite on every finite family of ordered lists of at most two points of S^{n-1}. Points may repeat, both within a list and across lists. A one-point list (y) is evaluated as the pair (y, y), as in the verification program. The kernels $\hat{Z}_\lambda$ are built from the zonal polynomial files computed with the program of [14]. In [13] the zonal matrices are Gram matrices, $Z_\lambda(J_1, J_2)_{\ell_1,\ell_2} = \langle \psi_{\lambda,\ell_1}(J_1), \psi_{\lambda,\ell_2}(J_2)\rangle$, so $\mathrm{ZonalPSD}(n, d_1, d_2)$ holds if the zonal files express these matrices correctly. The correctness of the zonal files is not formalized. In Lean we prove weak duality for the second level of the hierarchy, the nonnegativity of the multipliers $c\,g$ on Gram matrices of codes, and the bookkeeping that turns sums over pairs J_1, J_2 into the identities checked by the verification program.

There are two versions of the finite checks on a certificate (positive semidefiniteness of every matrix, the multiplier conditions, the four identities, and a symmetry of the zonal data). In the

first version, the checks are decided on the exact certificates by compiled code through Lean's `native_decide`, so the Lean compiler is part of the trusted base. This version builds in about five hours of wall time. In the second version, every check is done by the Lean kernel, and the six theorems depend only on Lean's standard axioms. This version uses the certificate matrices rounded to 30 decimal places, so that the kernel computations stay small. Lean checks that the rounded matrices are still positive semidefinite, through a diagonally dominant decomposition. The rounded matrices satisfy the identities only up to explicit residual polynomials r_k (stored in Lean as integer polynomials $L_k r_k$, where the positive integers L_k clear the denominators). Lean also proves a version of weak duality with residuals: $|C| \le K(\emptyset,\emptyset) + \sum_{k=1}^{4} \binom{|C|}{k} \|r_k\|_1$ for every code C, where $\|r\|_1$ is the sum of the absolute values of the coefficients of r. The bound on the residual terms uses only that the inner products of unit vectors lie in $[-1,1]$. If a code had more than M points, applying the inequality to $M+1$ of them would give $M+1 \le K(\emptyset,\emptyset) + \sum_k \binom{M+1}{k} \|r_k\|_1$. For the certificates of Theorem 1.1, with $M = 43$, 76 and 132, the right-hand side is less than $M+1$: $K(\emptyset,\emptyset)$ is $\frac{4399}{100}$, $\frac{7699}{100}$ and $\frac{13299}{100}$; the residual terms are about $2 \cdot 10^{-9}$, $4 \cdot 10^{-9}$ and $9 \cdot 10^{-9}$. In Lean's kernel, the checks pack vectors of integers into single natural numbers, so that most of the work is done by the kernel's built-in arithmetic on large numbers. The checks are split into pieces that fit in memory. For the certificates of Theorem 1.1 these checks take about 78, 77 and 76 hours, summed over the single-threaded pieces. The largest single check verifies positive semidefiniteness of the largest block (of size 1040). This check takes 7 to 8 hours and about 330 GB of memory for each of the three certificates.

To check ZonalPSD(n, d_1, d_2) numerically, we evaluated the kernels on random families of subsets of S^{n-1} (in total 1840, 1104 and 960 block matrices for the three certificates with $\cos\theta = \frac{1}{3}$; 960 for each certificate of Theorem 1.1). The smallest eigenvalue relative to $\max(1, \lambda_{\max})$ was never below -10^{-13}. In control runs on further families, we multiplied a single zonal coefficient by -3, and this change produced clearly negative eigenvalues in every family.

Data availability

The supplement [26] contains the certificates for Theorem 1.1 (in the file format of the verification program of [14]), the zonal polynomial files used by these certificates, the Lean files of Section 9, and the code that produced both the certificates and the zonal polynomial files. The supplement [27] contains the certificates for Theorem 1.2, the codes of Section 7 and a script that checks the codes in exact arithmetic.

Acknowledgements

Generative AI tools were used to write code, run analyses and draft text; the author directed the work and takes full responsibility for its content.

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